A Machine Learning-Based Tool for Autism Screening Using Psychological Medical Records

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Abstract. Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that impacts social interaction and behavior, where early detection is crucial for optimizing intervention outcomes.
This study proposes a machine learning-based screening tool for Autism Spectrum Disorder (ASD) using psychological electronic medical records (EMR) for children aged 1 - 8 years. Two datasets were utilized: (1) a public dataset with approximately 300 records and (2) a clinical EMR dataset with approximately 600 records. Both were labeled by psychology experts based on 18 behavioral evaluation criteria, including eye contact, pointing, and imitation.
Ten significant features were selected as inputs for training four machine learning models: Decision Tree, XGBoost, CatBoost, and Overall Local Accuracy (OLA). OLA demonstrated superior adaptability but faced challenges with class imbalance (90% ASD) and feature bias due to high-accuracy EMR data.
To address these, SMOTE-IPF was applied to balance the data, and OLA was enhanced with dynamic weighting. The proposed system shows high feasibility for ASD screening.

Keywords: Autism Spectrum Disorder, Machine Learning, Psychological Medical Records, SMOTE-IPF, OLA, Screening Tool

1 Introduction

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition marked by difficulties in social interaction, communication, and repetitive behaviors [?]. Early detection is vital for effective interventions, yet diagnosis becomes more challenging with age due to symptom overlap with other disorders. In the U.S., ASD prevalence has increased from 1 in 150 (2000) to 1 in 36 (2023) [?]; in Vietnam, estimates suggest 1 in 68 children, though official data remains limited.
Conventional tools like M-CHAT-R/F and ADOS require expert administration and are less accessible in low-resource settings such as Vietnam [?]. The absence of definitive medical tests further complicates diagnosis, relying on subjective behavioral assessments. Recent advances

in AI and ML enable automated ASD screening using electronic medical records (EMRs), though challenges like imbalanced data and feature bias remain [?]. While models like Decision Trees, XGBoost, and CatBoost perform well in controlled environments, they often lack generalizability in real-world data.

This study introduces an ML-based ASD screening system for children aged 1–8, using a combined dataset of 300 public and 600 clinical EMRs. We tackle class imbalance with SMOTE-IPF and improve prediction fairness via an enhanced Overall Local Accuracy (OLA) framework. Key contributions include: (1) a curated dataset with expert-annotated features; (2) a bias-reduced OLA model; and (3) a scalable, cost-effective solution for under-resourced environments.

The paper is structured as follows: Section 2 reviews related work; Section 3 describes the data and preprocessing; Section 4 outlines the methodology; Section 5 presents results; and Section 6 discusses contributions, limitations, and future directions.

2 Related Works

Harshita Chandrappa [?] developed a hybrid ML framework for ASD prediction across all age groups, integrating classifiers like Decision Tree, SVM, and Random Forest into a web-based diagnostic tool. SVM and Random Forest achieved the highest accuracy. The system also includes a chatbot interface to support early ASD awareness and symptom understanding.

Jose A. Saez [?] proposed SMOTE-IDF, a hybrid method combining SMOTE and IDF weighting to address class imbalance in text classification. Using a DNN, the approach preserved rare term significance and outperformed traditional methods on datasets like 20 Newsgroups in multiple evaluation metrics.

Youngkyu Hong [?] proposed a dual-branch learning framework to address bias in image classification. It uses Bias-Contrastive Learning (BCL) to capture spurious correlations in a biased branch and Bias-Balanced Learning (BBL) to focus on robust features in an unbiased branch. By minimizing biased branch loss and maximizing unbiased branch agreement with true labels, the method reduces reliance on dataset biases, outperforming traditional debiasing techniques on datasets like Biased-MNIST and ImageNet-9.

Y. Zeng [?] introduces Ola, an advanced omni-modal language model capable of understanding images, audio, and video. It uses progressive modality alignment-gradually learning from image-text, audio, then video-to reduce training costs. Ola supports streaming speech generation and demonstrates state-of-the-art performance among open-source omni-modal models, rivaling specialized systems across diverse tasks.

3 Dataset

This study utilizes two datasets to develop and evaluate an ASD detection approach:

Public Dataset

The public dataset was sourced from Kaggle [**?**], containing 292 records with balanced class distribution (48.3% ASD). Each record includes binary responses to 10 behavioral screening questions, personal information, and other relevant fields (20 columns total). These questions align with expert-defined evaluation criteria and were used to support consistent labeling of the private dataset.

Private Dataset

The private dataset consists of 594 anonymized medical records collected from a pediatric psychology clinic. Initially derived from over 1,200 psychological records covering various developmental conditions, the data was filtered and standardized through three steps:

- **Step 1:** Raw data was extracted from PDF documents and categorized by diagnosis, selecting over 800 ASD-related and non-clinical cases.
- **Step 2:** Incomplete or low-quality records were excluded, yielding 600 usable samples. Key behavioral features - such as eye contact [**?**], pointing [**?**], response to name [**?**], joint attention [**?**], imitation [**?**], and symbolic play [**?**] - were labeled under expert guidance. All personal identifiers were encoded to preserve privacy.
- **Step 3:** The final dataset contains 594 structured records, with 36 features (raw + labeled). The dataset is imbalanced, with 93

Together, these datasets provide a comprehensive foundation for building and validating a machine learning model for early ASD screening in children aged 1 - 8 years.
These datasets have 18 labeled features that are used for analysis and predict. These labeled features are listed below:

Table 1: List of labeled features in the dataset

Feature 1	**Feature 2**	**Feature 3**
Medical History	Task Requesting	Symbolic Play
Speech Delay	Response to Name in Infants	Repetitive/Stereotyped Play
Behavioral Rigidity	Toe Walking after 3 years	Social Communication Skills
Behavioral Isolation	Making a Point	Turn-Taking Play
Sensorimotor Play	Joint Attention	Imitation
Functional Play	Eye Contact	Combinatorial Play

4 Proposed Methodology

Below is a image and detail for each implementation method:

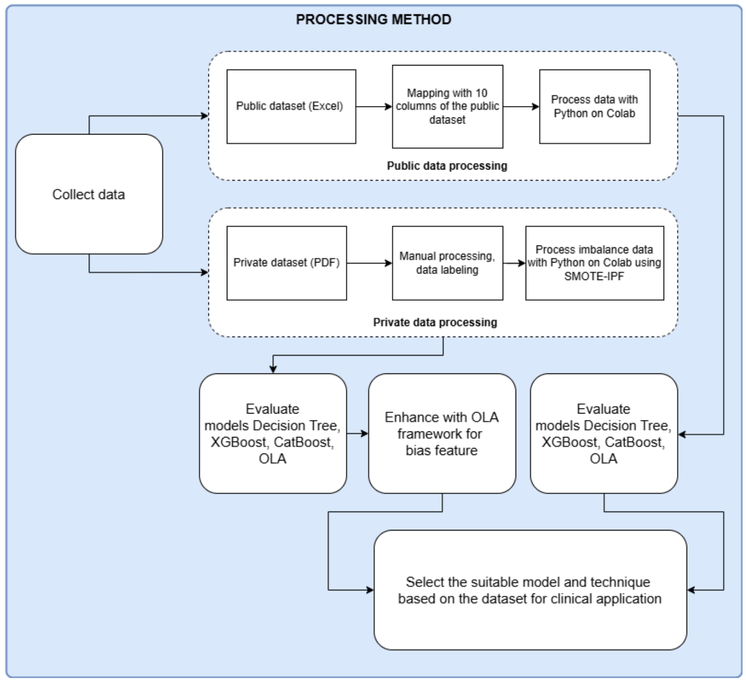


Fig. 1: Overview of implementation methods

4.1 Data Preprocessing

Effective data preprocessing is crucial to ensure the quality and usability of electronic medical records (EMR) for developing an Autism Spectrum Disorder (ASD) screening tool. The pipeline comprised five sequential stages, each with a clear *objective* and *method*, as detailed below.

Step 1: Anonymization All personally-identifiable information was either encrypted or removed before model training.

Step 2: Redundant Column Removal Columns lacking analytical value-such as purely administrative metadata-were eliminated.

Step 3: Missing Value Imputation Numerical attributes were imputed with the median, while categorical attributes were filled with the mode; domain-expert psychology rules guided these choices where appropriate.

Step 4: Feature Encoding

- Target variable (ASD diagnosis) was encoded as binary - `0` (non-ASD) and `1` (ASD).

– Behavioral features were encoded as binary (`0/1`) or ordinal values (`0`, `0.5`, `1`).

Step 5: SMOTE-IPF Balancing SMOTE-IPF [**?**] was selected as the main balancing method due to its effectiveness in handling class imbalance and filtering out noisy data. Unlike SMOTE-KNN [**?**], which imputes missing values but lacks noise filtering, SMOTE-IPF reduces the risk of overfitting.
To improve model training, unsafe ASD samples were further resampled using KNN, resulting in three datasets:
– A0: After SMOTE-IPF.
– A1: Fully amplified unsafe ASD samples.
– A2: Selective amplification based on neighbor voting.
These datasets, exported for training, showed that SMOTE-IPF offered the best trade-off between balance and robustness.

4.2 Evaluate Models

a, Models

Decision Tree A hierarchical structure of binary decisions based on feature thresholds to classify samples as ASD or non-ASD. Its simplicity and interpretability make it suitable for clinical applications. However, Decision Trees are prone to overfitting, especially on imbalanced datasets, and may overly rely on dominant features, such as specific behavioral indicators, leading to biased predictions. The model was implemented with default parameters, using Gini impurity as the splitting criterion, and evaluated on the preprocessed dataset. Additionally, its performance benefits from pruning techniques to reduce complexity, though it struggles with noisy data and lacks robustness against feature interactions, necessitating careful preprocessing to enhance accuracy in ASD classification.

XGBoost An ensemble learning method based on gradient boosting, XGBoost integrates multiple weak learners (decision trees) to enhance predictive performance. It effectively manages complex feature interactions and employs regularization to reduce overfitting. However, in imbalanced datasets, XGBoost may favor dominant features, potentially reducing its generalizability in pediatric ASD classification. Additionally, its efficiency stems from parallel processing and optimized tree pruning, though it requires careful hyperparameter tuning to address bias and improve adaptability across diverse data distributions.

CatBoost CatBoost is an advanced Gradient Boosting Decision Trees (GBDT) algorithm that builds an ensemble of symmetric (oblivious) decision trees to enhance predictive performance and computational efficiency. Unlike traditional Decision Trees, CatBoost combines multiple weak learners with ordered boosting and regularization to reduce overfitting, achieving superior generalization on complex datasets. It was configured with a depth of 6 and 100 iterations, leveraging its ability to process ordinal features (e.g., poor-moderate-good encodings) directly. However, CatBoost may still struggle with bias toward dominant features in imbalanced datasets, as applied in this study.

OLA (Overall Local Accuracy) [?] Overall Local Accuracy (OLA) [**?**] is a dynamic classifier selection method that chooses the most accurate base classifier for each test sample based on its performance within the sample's local neighborhood. For a given input, it finds the k-nearest neighbors in the training set and selects the classifier with the highest local accuracy to make the prediction.
In this study, OLA was applied to an ensemble of diverse classifiers (XGBoost, CatBoost and Random Forest). By adapting to local data patterns, OLA enhances prediction fairness and accuracy in ASD classification.

b, Evaluation Metrics and Validation

Evaluation Metrics Recall, Specificity, Precision, F1-score
Prioritizing the Recall (sensitivity) metric is driven by the primary goal of maximizing the detection of ASD cases, even if it results in some false negatives. This is particularly critical because missing a diagnosis of ASD in a child can lead to a lack of early intervention, potentially causing severe long-term developmental impacts. Recall measures the proportion of actual ASD cases correctly identified by the model, making it a preferred metric over Precision, F1-score. Even the low Specificity but high Recall is still rated as a suitable model.

Validation Techniques

Cross-Validation A 5-fold stratified cross-validation was conducted.
Process: The dataset was divided into 5 folds with preserved class distribution. Each fold was used once as test while others served as training.

Feature Importance Analysis Evaluates the influence of original features on the model's predictions, providing insights into feature relevance for the ASD classification task.

4.3 Enhance with OLA Framework

Framework with OLA This subsection illustrates the enhanced OLA pipeline designed to mitigate feature bias, particularly caused by dominant features such as `TiepXucMat`

- **Step 1:** Data Preprocessing
  - Standardization with StandardScaler: Transform data to a standard normal distribution (mean=0, variance=1) to reduce the influence of features with large values (eg: TiepXucMat), ensuring efficient model performance.

  - Dimensionality Reduction with PCA (Principal Component Analysis): Reduce data dimensions, removing noise and bias from dominant features while lowering computational cost.

- **Step 2:** Choosing internal model

  - Model Group Creation: XGBClassifier, CatBoostClassifier, and RandomForestClassifier trained on PCA data, focusing on key features to minimize noise.
  - This diversity enhances flexibility, allowing OLA to select the most suitable model for each test sample.

- **Step 3:** Deploy OLA Model
  Implement with the above models on the PCA-processed dataset to provide predictions for each data sample.

5 Result and Discussion

5.1 Experimental Results

Public Dataset

a. Metrics Cross Validation: Observations on Performance Metrics

- The Decision Tree model yields moderately low results, performing the least effectively among the evaluated models.

- CatBoost delivers good performance, though it is slightly outperformed by XGBoost, with a minimal difference of approximately 0.01.

- XGBoost exhibits very strong results, demonstrating high effectiveness.

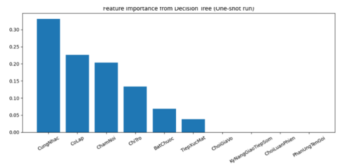
- Overall Local Accuracy (OLA) model also achieves very good performance, showcasing its adaptability to the data and surpassing both XGBoost and CatBoost in overall outcomes.

General comment, the Online Learning Algorithm (OLA) demonstrates superior effectiveness, with exceptional adaptability and high accuracy, positioning it as a highly promising model for deployment and further development on real-world datasets. XGBoost and CatBoost are also robust choices, with XGBoost slightly outperforming CatBoost by a marginal difference, indicating their strength in handling complex predictive tasks. Conversely, the Decision Tree model exhibits the lowest performance, making it more suitable for simpler problems or scenarios where model interpretability is prioritized over predictive power.

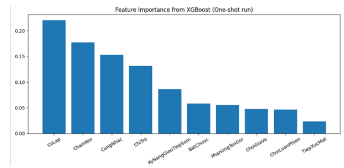
Table 2: Cross-validation performance (Public dataset)

Model	**Fold**	**Recall**	**Specificity**	**Precision**	**F1-Score**
Decision Tree	1	0.895	0.714	0.680	0.773
	2	0.842	0.893	0.842	0.842
	3	0.789	0.714	0.842	0.714
	4	0.889	0.857	0.800	0.842
	5	0.684	0.893	0.842	0.743
	Mean	**0.820**	**0.813**	**0.757**	**0.783**
XGBoost	1	1.000	0.929	0.957	0.950
	2	0.947	0.964	0.947	0.947
	3	1.000	1.000	1.000	1.000
	4	1.000	0.857	0.818	0.900
	5	0.947	1.000	1.000	0.973
	Mean	**0.979**	**0.950**	**0.944**	**0.954**
CatBoost	1	1.000	0.857	0.826	0.905
	2	1.000	0.964	0.950	0.974
	3	0.895	1.000	1.000	0.944
	4	1.000	0.893	0.857	0.923
	5	0.947	0.963	0.947	0.947
	Mean	**0.968**	**0.935**	**0.916**	**0.939**
OLA	1	1.000	1.000	1.000	1.000
	2	1.000	0.964	0.959	0.979
	3	1.000	1.000	1.000	1.000
	4	1.000	0.964	0.947	0.973
	5	0.947	1.000	1.000	0.973
	Mean	**0.989**	**0.986**	**0.979**	**0.985**

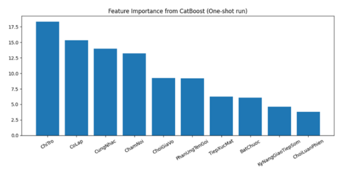
b. Chart Feature Importance Analysis: General comment, the chart results reveal that the dataset lacks significant column bias, resulting in a relatively uniform distribution across features. The models effectively recognize these features and produce graphs with balanced distributions, ranked from best to worst in a single run as follows: CatBoost - OLA - XGBoost - Decision Tree. OLA and CatBoost demonstrate superior column recognition compared to XGBoost and Decision Tree.



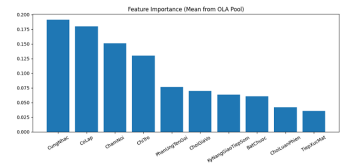
(a) Decision Tree



(b) XGBoost



(c) CatBoost



(d) OLA

Fig. 2: Baseline feature importance in public dataset

Private Dataset

a. Metrics Cross Validation: - The Decision Tree model maintains a moderate performance, consistent with its predictions on standardized datasets, but remains the least effective among the evaluated models.

- CatBoost, while demonstrating good performance, exhibits a notable decline compared to its predictions on standardized datasets. This reduction is attributed to the instability of the dataset and CatBoost's suitability for larger, more stable datasets.

- XGBoost, initially predicted to yield excellent results, achieves a good performance with a metric of 0.9, outperforming both Decision Tree and CatBoost, though it falls short of its anticipated excellence due to data variability.

General comment, the Overall Local Accuracy (OLA) delivers outstanding performance, surpassing other models on this real-world dataset. Despite a slight reduction in metrics compared to predictions (approximately 0.02), OLA's adaptability to the unstable data from the clinical setting is evident, outperforming XGBoost and CatBoost in flexibility. Notably, the OLA Framework shows no significant improvement over traditional OLA, with evaluation metrics remaining comparable or exhibiting minimal differences.

These findings suggest that OLA is the most suitable model for datasets lacking stability and clear trends, as encountered in this study. However, potential biases in the dataset's feature columns may contribute to the observed instability, warranting further investigation to mitigate such issues in future applications.

Table 3: Cross-validation performance (Private dataset)

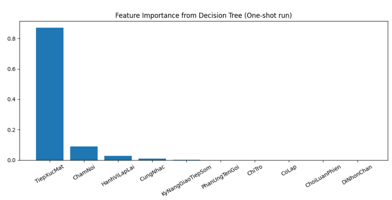
Model	Fold	Recall	Specificity	Precision	F1-Score
Decision Tree	1	0.920	0.820	0.836	0.876
	2	0.760	0.940	0.927	0.835
	3	0.840	0.840	0.840	0.840
	4	0.880	0.740	0.772	0.822
	5	0.740	0.840	0.822	0.779
	Mean	**0.828**	**0.836**	**0.839**	**0.830**
XGBoost	1	0.920	0.900	0.902	0.911
	2	0.940	0.960	0.959	0.949
	3	0.900	0.900	0.900	0.900
	4	0.960	0.840	0.857	0.906
	5	0.800	0.880	0.870	0.833
	Mean	**0.904**	**0.896**	**0.898**	**0.900**
CatBoost	1	0.900	0.880	0.882	0.891
	2	0.920	1.000	1.000	0.958
	3	0.920	0.920	0.918	0.909
	4	0.940	0.780	0.810	0.870
	5	0.820	0.880	0.872	0.845
	Mean	**0.896**	**0.892**	**0.896**	**0.895**
OLA	1	0.940	1.000	1.000	0.969
	2	0.980	1.000	1.000	0.990
	3	1.000	1.000	1.000	1.000
	4	0.980	0.980	0.980	0.980
	5	0.920	1.000	1.000	0.958
	Mean	**0.964**	**0.996**	**0.996**	**0.979**
Fw OLA	1	0.969	0.973	0.979	0.974
	2	0.980	0.973	0.980	0.980
	3	0.969	0.987	0.990	0.979
	4	0.980	0.987	0.990	0.985
	5	0.938	1.000	1.000	0.968
	Mean	**0.967**	**0.984**	**0.988**	**0.977**

b. Chart Feature Importance Analysis: The bias column evaluation charts reveal the following insights:

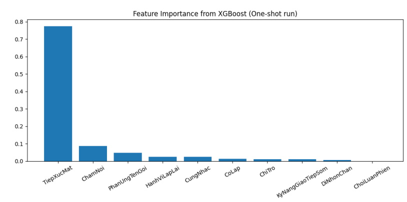
The dataset exhibits a significant bias toward the "TiepXucMat" (Eye Contact) column across all four models, leading to overly optimistic evaluation metrics. However, when applied to new datasets in real-world scenarios, the results fall short of expectations if the "TiepXucMat" column is not accurately assessed.

Although CatBoost shows lower Cross Validation performance compared to other models, it maintains a better distribution of the biased column. However, it cannot be conclusively regarded as the best model for column distribution.

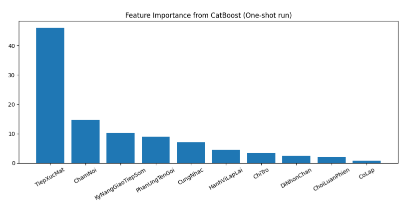
OLA remains a promising model for further improvement due to its flexible adaptability.
General comment, the dataset is predominantly biased toward the "TiepXucMat" column, necessitating enhancements to the adaptable OLA model. These improvements should aim to achieve a more uniform and balanced differentiation across columns, ensuring that the screening process does not miss positive cases.



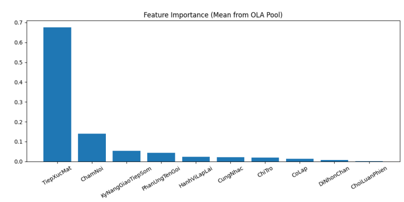
(a) Decision Tree



(b) XGBoost



(c) CatBoost



(d) Initial OLA

Fig. 3: Baseline feature importance in private dataset

To address this, Enhanced OLA was applied to reduce bias and diversify feature influence.

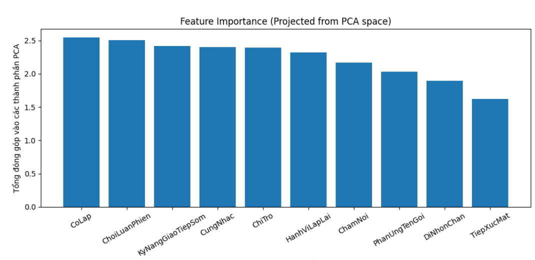


Fig. 4: Enhanced OLA framework with TiepXucMat

Summary: Enhanced OLA framework successfully reduced the dominance of *TiepXucMat*, improving balance among features and robustness for ASD classification on the private dataset.

5.2 Discussion

This section analyzes the performance patterns across the evaluated models, discusses the improvements of OLA's local selection mechanism over global bias, compares our results with related work, and highlights limitations, strengths, weaknesses, and the practical applicability of OLA on the private dataset.

Table 4: Compare Recall on public and private datasets

Model	**Public**	**Private**
Decision Tree	0.820	0.828
XGBoost	0.979	0.904
CatBoost	0.968	0.896
OLA	0.989	0.964
OLA Framework	–	0.967

Compare Recall on public and private datasets, as depicted in the provided table, highlight the performance of various machine learning models across public and private datasets, with a focus on Recall as the primary metric due to its importance in ensuring maximum detection of ASD cases. The table shows that OLA and the OLA Framework in practical data achieved the highest Recall scores of 0.964 and 0.967, respectively, significantly outperforming CatBoost (0.896), XGBoost (0.904), and Decision Tree (0.828). This superior performance underscores OLA's exceptional flexibility in dynamically selecting optimal models, reinforcing its robustness and adaptability across diverse datasets.
A notable observation is the decline in accuracy when these models are applied to the private clinical dataset compared to the public dataset. XGBoost dropped from 0.979 to 0.904, CatBoost from 0.968 to 0.896, and OLA from 0.989 to 0.964-a reduction of only 0.025. This decrease indicates that while all models perform well (above 0.9 on average), their effectiveness diminishes in real-world settings with unstable data, failing to replicate the high accuracy observed in controlled experiments. The private dataset's variability reflects the challenges of real-world clinical environments, where data inconsistencies are common.
Further analysis of column bias reveals a significant skew toward the "TiepXucMat" (Eye Contact) column in the private dataset, which likely contributed to the overly optimistic results in initial evaluations. This bias suggests that the models' predictions may overly rely on a single feature, potentially leading to inaccurate assessments when applied practically. The OLA Framework was introduced to address this limitation, balancing feature weights and integrating the 10-question framework across 10 screening themes. Post-implementation, the OLA Framework successfully mitigated this bias, still achieving a Recall of 0.967, demonstrating its ability to adapt to real-world data fluctuations.

[0] OLA Framework doesn't need to be assessed on public dataset so "–".

6 Conclusion

This study introduces a screening tool leveraging machine learning models trained on psychological medical records of children, utilizing the Overall Local Accuracy (OLA) framework following advanced preprocessing techniques. These include balancing class labels with SMOTE-IPF and mitigating column bias through the OLA framework. Traditional models, including advanced ones like CatBoost, struggled with imbalanced data, facing limitations due to a "column causing data bias" that undermined reliability, particularly given the hidden biases often present in real-world datasets. The enhanced OLA, equipped with additional techniques to control this bias, proved to be a robust solution, significantly improving ASD case detection and ensuring stable performance. The primary contribution of this research is a novel approach to developing an ASD screening tool, with potential applicability to screening other medical conditions using psychological medical records and machine learning. This work lays the groundwork for more accurate, equitable, and user-friendly diagnostic tools, particularly for early screening where missing cases are a critical concern. However, challenges remain, including the limited dataset size and the unreliable nature of manual labeling. Future efforts should focus on testing the OLA framework on larger datasets, validated through real-world software trials combined with corresponding medical records to further train the model. This approach will ensure the capture of outlier cases and enable broader scalability, extending the tool's use beyond a single clinic to multiple healthcare facilities and hospitals nationwide. The OLA Framework's strength lies in its dynamic model selection and adaptability, making it highly suitable for building ASD screening tools in the context of volatile real-world datasets. Its ability to maintain high performance despite data instability and column bias positions it as an ideal solution for clinical applications, especially in resource-constrained settings where robust and flexible tools are vital. Future enhancements should prioritize refining feature weighting to improve generalizability across diverse populations.

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References

1. Holly Hodges, Casey Fealko, Neelkamal Soares. "Autism spectrum disorder: definition, epidemiology, causes, and clinical evaluation", 2020. doi: https://doi.org/10.21037/tp.2019.09.09
2. Bang Le. "Ty le tre tu ky o Viet Nam va The gioi: Nhung con so bao dong", 2025. url: https://tretuky.edu.vn/ty-le-tre-tu-ky-o-viet-nam-va-the-gioi/
3. Clara Lucato dos Santos, Indyanara Inacio Barreto, Idevaldo Floriano, Luca Schiliró Tristao, Antonio Silvinato, Wanderley Marques Bernardo. "Screening and diagnostic tools for autism spectrum disorder: Systematic review and meta-analysis", March 2024. doi: https://doi.org/10.1016/j.clinsp.2023.100323
4. Paul Olubudo. "Statistical Bias Correction in Imbalanced Datasets", April 2025. doi: https://www.researchgate.net/publication/390805682
5. Harshita Chandrappa, Swaathi BR, Prof. Swimpy Pahuja. "Prediction of Autism Spectrum Disorder based on Machine Learning Approach," *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, vol. 11, no. V, pp. 3521-3526, May 2023. doi: 10.22214/ijraset.2023.52417. Available: `https://doi.org/10.22214/ijraset.2023.52417`
6. J. A. Sáez, J. Luengo, J. Stefanowski, and F. Herrera, "A novel technique for text classification using SMOTE-IPF based oversampling and deep neural network" *Int. J. of Advanced Computer Science and Applications*, vol. 12, no. 10, 2014. doi: 10.14569/IJACSA.2021.051.
7. H. Karamti *et al.*, "SMOTE-KNN: An improved algorithm based on SMOTE and K-nearest neighbors for imbalanced data," *IEEE Access*, vol. 9, pp. 95690–95701, 2023. doi: 10.1109/ACCESS.2023.5174412.
8. B. D. Hung, V. V. Thoa, and X. T. Dang, "KSI - A combined clustering and resampling method with noise filtering algorithm for imbalanced data classification," *J. of Information and Communication Technology*, vol. 1, no. 1, pp. xx–xx, 2019.
9. Y. Hong and E. Yang, "Unbiased classification through bias-contrastive and bias-balanced learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, pp. 20153–20165, 2021.
10. Z. Liu *et al.*, "Ola: Pushing the frontiers of omni-modal language model with progressive modality alignment," *arXiv preprint*, arXiv:2502.04328, 2025. Available: `https://arxiv.org/abs/2502.04328`
11. J. M. Moriuchi, A. Klin, and W. Jones, "Mechanisms of diminished attention to eyes in autism," *Am. J. of Psychiatry*, vol. 173, no. 8, pp. 825–833, 2016. doi: 10.1176/appi.ajp.2016.15081034.
12. S. Ramos-Cabo, V. Vulchanov, and M. Vulchanova, "Different ways of making a point: A study of gestural communication in typical and atypical early development," *Autism Research*, vol. 14, no. 5, pp. 984–996, 2021. doi: 10.1002/aur.2474.

13. M. Miller *et al.*, "Response to name in infants developing autism spectrum disorder: A prospective study," *The J. of Pediatrics*, vol. 165, no. 2, pp. 332–337, 2014. doi: 10.1016/j.jpeds.2014.04.017.
14. M. Montagut-Asunción *et al.*, "Joint attention and its relationship with autism risk markers at 18 months of age," *Children*, vol. 9, no. 4, 2022. doi: 10.3390/children9040480.
15. B. Ingersoll, "The social role of imitation in autism: Implications for the treatment of imitation deficits," *Infants & Young Children*, vol. 21, no. 2, pp. 107–119, 2008. doi: 10.1097/01.IYC.0000314482.24087.14.
16. Y. G. Lam and S. S. Yeung, "Symbolic play in children with autism," in *Comprehensive Guide to Autism*, Springer, 2014, pp. 491–508. doi: 10.1007/978-1-4614-4788-7_26.
17. F. F. Tabtah, "Autism spectrum disorder screening," dataset about list 10 question to predict ASD in kaggle in *Proc. 1st Int. Conf. on Medical and Health Informatics*, Taichung City, Taiwan, ACM, 2017. Available: `http://fadifayez.com/wp-content/uploads/2017/11/Autism-Spectrum-Disorder-Screening-Machine-Learning.pdf`
18. Overall Local Accuracy. Available: `https://deslib.readthedocs.io/en/latest/modules/dcs/ola.html`
19. R. M. O. Cruz, R. Sabourin, and G. D. C. Cavalcanti, "DESlib: A dynamic ensemble selection library in Python," 2018. Available: `https://github.com/scikit-learn-contrib/DESlib`